Quick Select:

Data: [157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133]

K = 3.

Initial array:

[157, 110, 147, 122, 111, 149, 157, 141, 123, 112, 117, 133]

Choose pivot

pivot = 133 (last element)

Reorder array based on

elements < 133 to the left of 133

elements > 133 to the right of 133

After partition:

[110, 122, 111, 123, 112, 117, 133, 141, 157, 151, 149]

Pivot index = 6

pivot index > k [6 > 2] [index of k is 2

As 6 > 2

we should consider left subarray

Now,

[110, 122, 111, 123, 112, 117]

new pivot = 117 (last element)

After partition:

[110, 111, 112, 117, 122, 123]

pivot index = 3

pivot index > k index [3 > 2]

Now → [110, 111, 112]

pivot index = k [2 = 2]

k^{th} smallest number = 112.